Unit 3 End-of-Unit Assessment - Form A Teacher Key

8 items · 3 points each · 24 total. Award 1 for a valid model/setup, 1 for correct mathematics/units, and 1 for a requested justification. Use the applicable evidence when the task is conceptual; accept equivalent methods and response modes. Do not count writing length as mathematical mastery.

Item 1

Prompt: There are 8 blue and 12 yellow counters. Give blue:yellow and blue:all in simplest form, explaining the whole.

Solution / evidence: 2:3 and 2:5; all includes both colors.

Item 2

Prompt: A machine makes 48 pages in 4 minutes at a steady rate. Give unit rate and output in 5 minutes.

Solution / evidence: 12 pages/minute; 60 pages.

Item 3

Prompt: A pack of 4 pens costs $3.2; a pack of 6 costs $4.5. Compare unit prices.

Solution / evidence: $0.80 versus $0.75 per pen; second pack has lower unit price.

Item 4

Prompt: A mixture is 4:6. Find the second amount when the first is 8. Show equivalent scaling.

Solution / evidence: 12; factor 2 or bridge 2:3.

Item 5

Prompt: A rate is $3 per kilogram. Give the point for 4 kg and interpret its coordinates.

Solution / evidence: (4,12); 4 kilograms cost $12.

Item 6

Prompt: Mix A uses 2 parts concentrate to 5 water; B uses 3 to 7. Compare with a common concentrate amount.

Solution / evidence: At 6 concentrate, A has 15 water and B 14. B is stronger under the stated ratio comparison; compare less water for equal concentrate, or state equal strength if equal.

Item 7

Prompt: A project needs 13 feet of ribbon. How many whole yards must be bought? Use 1 yd=3 ft and test the smaller purchase.

Solution / evidence: 5 yards; 4 yards is only 12 feet and is insufficient.

Item 8

Prompt: Use 1 kg≈2.2 lb. Convert 8.8 lb to kilograms and explain why the conversion is approximate.

Solution / evidence: Approximately 4 kg; the supplied factor is rounded. Reverse check 4×2.2=8.8.

Response: group errors by interpretation/model versus calculation versus explanation. Use the corresponding lesson Teaching Support Guide, revise one step, and reassess with its reserved fresh check. Do not reuse the already seen form as independent evidence.